PBB Bridge B03 — Expanded Metrics Catalog (Addendum to v2 Guide)

Purpose: the v2 guide used ~15-20 of 320 inputs, mostly lumped into two categories (Safety, Equipment-failure). This addendum mines all 9 signal categories from the IO4 guide into per-subsystem metrics, and slots each into the tab where it belongs. Use alongside `PBB_Report_Metrics_UX_Guide_v2.md` — this doesn't replace that structure, it fills it in.

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1. Equipment failures — break the lump into subsystems

Currently: one "Equipment-failure faults/day" line. The category actually spans four distinct subsystems (inputs 53–67, 87–93, 116–117, 120, 131–133, 319):

New metric	Inputs	Tab
Engine fault count/day	subset of 53–67	Daily Trends, Bridge Detail → Maintenance
Analog/sensor fault count/day	subset of 53–67, 87–93	Daily Trends, Maintenance
PLC fault count/day	116–117, 120	Daily Trends, Maintenance
Joystick fault count/day	131–133	Daily Trends, Maintenance
Anticollision comms fault count/day	319	Daily Trends, Maintenance
Subsystem fault breakdown table (7d/30d count, % of total, last occurred)	all of the above	Bridge Detail → Maintenance (replaces single aggregate row)

This turns one flat number into a table that tells ops *which part of the bridge* is failing — directly actionable, and it's exactly the kind of subsystem breakdown the reference report gives for jerk/vibration by phase (start/cruise/stop).

2. Autolevelling & docking — more than a timeout count

Currently: “Autolevelling activations/day” only. Inputs 48–49, 67, 72–80, 125–127, 303 support more:

New metric	Tab
Autolevel timeout count/day (distinct from total activations)	Daily Trends
Wrong-docking flag count/day	Daily Trends, Anomalous Events
Safety-shoe-in-auto-mode trigger count	Daily Trends
Autolevel success rate (%) — activations that complete without timeout	Operational Health

3. Position limits — reframe from “noise” to wear tracking

Currently excluded as “often transient.” That’s correct for alerting, but the *frequency* of hitting each limit is a legitimate wear/calibration signal even without vibration data — this is a real way to approximate mechanical condition monitoring using only discrete bits.

New metric	Inputs	Tab
Height limit hits/day	subset 11–15, 145–156	Daily Trends
Slope limit hits/day	subset 145–156	Daily Trends
Extension limit hits/day	subset 174–175, 218–221	Daily Trends
Rotunda limit hits/day	subset 254–265	Daily Trends
Bridgehead limit hits/day	subset 254–265	Daily Trends

Limit-hit trend flag — “rotunda limit hits up 35%/week — possible drift, recommend calibration check”	AI Maintenance Recommendations

Caveat to carry into the UI: these are *not* faults on their own — frame explicitly as “movement pattern” metrics, not alerts, so ops doesn’t mistake normal operation for a problem. A rising trend is the signal, not the raw count.

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4. Ground support (GPU / PCAir / 400Hz) — this maps straight to the original ROI ask

Currently: only “connection status” (on/off) and a single “connect duration/day” number. Inputs 99, 101, 160–167, 277–282 support real utilization metrics:

New metric	Tab
GPU connect duration/day	Operational Health, Daily Trends
PCAir active duration/day	Operational Health, Daily Trends
400Hz active duration/day	Operational Health, Daily Trends
Ground support alarm count/day (per umbilical)	Daily Trends, Anomalous Events
% of dockings with full ground-support hookup (all three connected) vs. partial	Executive Summary, Operational Health
Avg time-to-hookup after docking (GPU/PCAir connect timestamp – docked timestamp)	Operational Health

That last one is genuinely valuable to an airport ops manager — it measures ground crew responsiveness, not the bridge itself, but it’s visible in this data and nobody else is tracking it.

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5. Anticollision & wing protection — safety-adjacent, currently invisible

Currently: not surfaced at all. Inputs 168, 170–171, 213, 275–276, 286:

New metric	Tab
Anticollision trigger count/day	Daily Trends
Wing-detection trigger count/day	Daily Trends
Anticollision comms dropout events	Anomalous Events (already have the fault count; this is the *comms* health specifically)

These are near-miss proxies — worth a light callout in Executive Summary if trigger rate rises, since it's the closest thing to a safety leading-indicator this device offers outside the Safety category itself.

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6. Automation adoption — the metric that answers the original ROI question

The very first reference deck was titled *"Remote/Automatic Operation — Data to build user case and develop ROI."* We've been using input 284 (Point-and-Go) only as a raw activation count. It deserves its own section:

New metric	Tab
% of dockings using Point-and-Go (automated) vs. manual	Executive Summary, Operational Health
Automation adoption trend (%/week)	Daily Trends
Avg docking duration — automated vs. manual (comparison)	Operational Health
Manual-override rate (dockings started automated, finished manual)	Anomalous Events

If automated dockings are measurably faster or more consistent than manual ones, this is the single most direct "ROI" statement the whole dashboard can make — worth prioritizing.

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7. Stand configuration — static context, not a trend

Inputs 272–273 (triple/double stand) don't change often. Don't build a trend chart for this — just surface it as a static fact in Bridge Detail → Settings ("Stand type: Double"), same treatment as OEM/model info.

Where this lands in the existing tab structure

Tab	What gets added
Executive Summary	Subsystem mini-indicators (Safety / Equipment / Ground support / Anticollision) instead of one lumped condition sentence; automation adoption %
Operational Health	Ground support utilization, autolevel success rate, automated-vs-manual duration comparison
AI Maintenance Recommendations	Per-subsystem recommendations (engine/PLC/joystick/anticollision-comms) instead of one generic equipment-failure line; limit-drift flags
Daily Trends	Expands from 2 category lines to ~12 subsystem-level trend rows (only showing ones that clear the significance bar, per v2 guide's existing rule)
Anomalous Events	Adds ground-support disconnect events, manual-override events, limit-hit bursts
Bridge Detail → Maintenance	Subsystem fault breakdown table replaces single aggregate
Bridge Detail → Settings	Stand configuration as static fact

One caution before handing this to Claude Design

More metrics is only useful if the *hierarchy* stays clear — the earlier "too little information" problem was real, but the risk on the other side is turning Daily Trends into 20 undifferentiated charts an ops manager has to scan every morning. Recommend keeping the **Executive Summary and Operational Health tabs deliberately sparse** (still the 5-8 headline numbers a manager scans daily) and letting the **subsystem-level depth live in Daily Trends, Anomalous Events, and Bridge Detail** — the tabs someone opens when something looks off, not the ones they check every morning.